

Windows PowerShell

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PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> dir

Directory: C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice

Mode	LastWriteTime	Length	Name
-a----	28-09-2026 13:02	65	.dockerignore
-a----	28-09-2026 13:02	6724	api.py
-a----	28-09-2026 13:02	907	common.py
-a----	28-09-2026 13:02	1157	compose.yaml
-a----	28-09-2026 13:02	437	consume_model.py
-a----	28-09-2026 13:02	207	Dockerfile.dev
-a----	28-09-2026 13:02	335	Dockerfile.service
-a----	28-09-2026 13:02	738	drift.py
-a----	28-09-2026 13:02	886	export_champion.py
-a----	28-09-2026 13:02	1474	generate_data.py
-a----	28-09-2026 13:02	8865	Lab3_Colab.ipynb
-a----	28-09-2026 13:02	3579	labctl.py
-a----	28-09-2026 13:02	1064	load_test.py
-a----	28-09-2026 13:02	1010	probe_service.py
-a----	28-09-2026 13:02	1368	register_model.py
-a----	28-09-2026 13:02	235	requirements.txt
-a----	28-09-2026 13:02	1917	run_lab1.py
-a----	28-09-2026 13:02	51	serve_churn.py
-a----	28-09-2026 13:02	7361	START_HERE.html
-a----	28-09-2026 13:02	7047	START_HERE.md

-a----	28-09-2026	13:02	3264 SUBMISSION_TEMPLATE.html
-a----	28-09-2026	13:02	551 train_auto.py
-a----	28-09-2026	13:02	562 train_no_tracking.py
-a----	28-09-2026	13:02	3125 train_tracked.py
-a----	28-09-2026	13:02	4259 VALIDATION.md
-a----	28-09-2026	13:02	3499 WRITTEN_ANSWERS.md

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker info

Client:

Version: 26.1.1

Context: default

Debug Mode: false

Plugins:

agent: Docker AI Agent Runner (Docker Inc.)

Version: v1.140.0

Path: C:\Users\saian\.docker\cli-plugins\docker-agent.exe

ai: Docker AI Agent - Ask Gordon (Docker Inc.)

Version: v1.31.0

Path: C:\Users\saian\.docker\cli-plugins\docker-ai.exe

buildx: Docker Buildx (Docker Inc.)

Version: v0.37.1

Path: C:\Users\saian\.docker\cli-plugins\docker-buildx.exe

compose: Docker Compose (Docker Inc.)

Version: v5.5.1

Path: C:\Users\saian\.docker\cli-plugins\docker-compose.exe

debug: Get a shell into any image or container (Docker Inc.)

Version: 0.0.47

Path: C:\Users\saian\.docker\cli-plugins\docker-debug.exe

desktop: Docker Desktop commands (Docker Inc.)

Version: v0.4.4

Path: C:\Users\saian\.docker\cli-plugins\docker-desktop.exe
dev: Docker Dev Environments (Docker Inc.)
Version: v0.1.2

Path: C:\Program Files\Docker\cli-plugins\docker-dev.exe
dhi: CLI for managing Docker Hardened Images (Docker Inc.)
Version: v0.0.7

Path: C:\Users\saian\.docker\cli-plugins\docker-dhi.exe
extension: Manages Docker extensions (Docker Inc.)
Version: v0.2.31

Path: C:\Users\saian\.docker\cli-plugins\docker-extension.exe
feedback: Provide feedback, right in your terminal! (Docker Inc.)
Version: v1.0.4

Path: C:\Program Files\Docker\cli-plugins\docker-feedback.exe
init: Creates Docker-related starter files for your project (Docker Inc.)
Version: v1.4.0

Path: C:\Users\saian\.docker\cli-plugins\docker-init.exe
mcp: Docker MCP Plugin (Docker Inc.)
Version: v0.43.3

Path: C:\Users\saian\.docker\cli-plugins\docker-mcp.exe
model: Docker Model Runner (Docker Inc.)
Version: v1.2.6

Path: C:\Users\saian\.docker\cli-plugins\docker-model.exe
offload: Docker Offload (Docker Inc.)
Version: v0.6.27

Path: C:\Users\saian\.docker\cli-plugins\docker-offload.exe
pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
Version: v0.2.2

Path: C:\Users\saian\.docker\cli-plugins\docker-pass.exe
sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
Version: v0.13.0

Path: C:\Users\saian\.docker\cli-plugins\docker-sandbox.exe

sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)

Version: 0.6.0

Path: C:\Program Files\Docker\cli-plugins\docker-sbom.exe

scout: Docker Scout (Docker Inc.)

Version: v1.24.0

Path: C:\Users\saian\.docker\cli-plugins\docker-scout.exe

Server:

Containers: 0

Running: 0

Paused: 0

Stopped: 0

Images: 7

Server Version: 26.1.1

Storage Driver: overlay2

Backing Filesystem: extfs

Supports d_type: true

Using metacopy: false

Native Overlay Diff: true

userxattr: false

Logging Driver: json-file

Cgroup Driver: cgroupfs

Cgroup Version: 2

Plugins:

Volume: local

Network: bridge host ipvlan macvlan null overlay

Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog

Swarm: inactive

Runtimes: runc io.containerd.runc.v2

Default Runtime: runc

Init Binary: docker-init

containerd version: e377cd56a71523140ca6ae87e30244719194a521

runc version: v1.1.12-0-g51d5e94

init version: de40ad0

Security Options:

seccomp

Profile: unconfined

cgroupns

Kernel Version: 6.18.33.2-microsoft-standard-WSL2

Operating System: Docker Desktop

OSType: linux

Architecture: x86_64

CPUs: 12

Total Memory: 3.674GiB

Name: docker-desktop

ID: e3776b19-fb4c-4c66-a67a-6e0076e43583

Docker Root Dir: /var/lib/docker

Debug Mode: false

HTTP Proxy: http.docker.internal:3128

HTTPS Proxy: http.docker.internal:3128

No Proxy: hubproxy.docker.internal

Labels:

com.docker.desktop.address=npipe:///\\.\pipe\docker_cli

Experimental: false

Insecure Registries:

hubproxy.docker.internal:5555

127.0.0.0/8

Live Restore Enabled: false

WARNING: daemon is not using the default seccomp profile

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice>

\$env:LAB_AUTHOR="Sai 2025AE05621_3"

```
PS C:\Users\saiant\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> python labctl.py  
init
```

```
[+] Building 4.5s (12/12) FINISHED
```

```
=> [internal] load local bake definitions 0.0s  
=> => reading from stdin 608B 0.0s  
=> [internal] load build definition from Dockerfile.dev 0.1s  
=> => transferring dockerfile: 250B 0.1s  
=> [internal] load metadata for docker.io/library/python:3.11-slim 2.7s  
=> [auth] library/python:pull token for registry-1.docker.io 0.0s  
=> [internal] load .dockerignore 0.0s  
=> => transferring context: 105B 0.0s  
=> [1/4] FROM docker.io/library/python:3.11-  
slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41 0.0s  
=> [internal] load build context 0.0s  
=> => transferring context: 280B 0.0s  
=> CACHED [2/4] WORKDIR /workspace 0.0s  
=> CACHED [3/4] COPY requirements.txt /tmp/requirements.txt 0.0s  
=> CACHED [4/4] RUN pip install --no-cache-dir -r /tmp/requirements.txt 0.0s  
=> exporting to image 0.1s  
=> => exporting layers 0.0s  
=> => writing image  
sha256:dd505a4518148d767a8290cc2d42b35c5f84acd1227ea3655143c7e2e51411ba  
0.0s  
=> => naming to docker.io/library/cartvista-dev:lab 0.0s  
=> resolving provenance for metadata file 0.1s
```

```
[+] build 1/1
```

```
✔ Image cartvista-dev:lab Built 4.9s
```

```
[+] run 2/2
```

```
✔ Volume cartvista_lab3_project Created 0.0s
```

```
✔ Network cartvista-lab3_default Created 0.2s
```

```
Container cartvista-lab3-bootstrap Creating
```

```
Container cartvista-lab3-bootstrap Created
```

860d8976b1c35ef2cfa9cf968b17e2e5482de95dd79a31b742a93421a723ab52

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/.dockerignore

Successfully copied 8.7kB to cartvista-lab3-bootstrap:/workspace/api.py

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/common.py

Successfully copied 3.07kB to cartvista-lab3-bootstrap:/workspace/compose.yaml

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/consume_model.py

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/Dockerfile.dev

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/Dockerfile.service

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/drift.py

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/export_champion.py

Successfully copied 3.07kB to cartvista-lab3-bootstrap:/workspace/generate_data.py

Successfully copied 10.8kB to cartvista-lab3-bootstrap:/workspace/Lab3_Colab.ipynb

Successfully copied 5.12kB to cartvista-lab3-bootstrap:/workspace/labctl.py

Successfully copied 3.07kB to cartvista-lab3-bootstrap:/workspace/load_test.py

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/probe_service.py

Successfully copied 3.07kB to cartvista-lab3-bootstrap:/workspace/register_model.py

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/requirements.txt

Successfully copied 3.58kB to cartvista-lab3-bootstrap:/workspace/run_lab1.py

Successfully copied 2.05kB to cartvista-lab3-bootstrap:/workspace/serve_churn.py

Successfully copied 9.22kB to cartvista-lab3-bootstrap:/workspace/START_HERE.html

Successfully copied 8.7kB to cartvista-lab3-bootstrap:/workspace/START_HERE.md

Successfully copied 5.12kB to cartvista-lab3-bootstrap:/workspace/SUBMISSION_TEMPLATE.html

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/train_auto.py

Successfully copied 2.56kB to cartvista-lab3-bootstrap:/workspace/train_no_tracking.py

Successfully copied 5.12kB to cartvista-lab3-bootstrap:/workspace/train_tracked.py

Successfully copied 6.14kB to cartvista-lab3-bootstrap:/workspace/VALIDATION.md

Successfully copied 5.12kB to cartvista-lab3-bootstrap:/workspace/WRITTEN_ANSWERS.md

cartvista-lab3-bootstrap

Initialized. Run python labctl.py prepare next. Do not repeat init on an active project.

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice>
\$SourceVolume="cartvista_lab2_project"

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker volume inspect $SourceVolume
```

```
[
  {
    "CreatedAt": "2026-09-28T10:53:07Z",
    "Driver": "local",
    "Labels": {
      "com.docker.compose.config-hash":
        "e2305a962e8b218a8962fc01a48d36da6dd6a41baf0bfd41d271bd501aebea9f",
      "com.docker.compose.project": "cartvista-lab2",
      "com.docker.compose.version": "5.5.1",
      "com.docker.compose.volume": "project"
    },
    "Mountpoint": "/var/lib/docker/volumes/cartvista_lab2_project/_data",
    "Name": "cartvista_lab2_project",
    "Options": null,
    "Scope": "local"
  }
]
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm --no-deps --user 0:0 -v "${SourceVolume}:/previous:ro" tools sh -c 'test -f
/previous/mlflow.db && test -d /previous/mlruns && test ! -e /workspace/mlflow.db && cp
/previous/mlflow.db* /workspace/ && cp /previous/churn_data.csv /workspace/ && cp -a
/previous/mlruns /workspace/ && chown -R 50000:0 /workspace/mlflow.db* /workspace/mlruns
/workspace/churn_data.csv'
```

Container cartvista-lab3-tools-run-6f4036cb118b Creating

Container cartvista-lab3-tools-run-6f4036cb118b Created

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm tools python consume_model.py
```

Container cartvista-lab3-tools-run-f1842582f8d3 Creating

Container cartvista-lab3-tools-run-f1842582f8d3 Created

```
{
  "version": 2,
  "run_id": "f0dc4382ba9840789fd4d27177ad45a5",
```

```
"predictions": [
  0,
  0,
  0,
  0,
  0
],
"probabilities": [
  0.0877304659981777,
  0.018408666625937913,
  0.06312212385713238,
  0.03516482773043705,
  0.11565878631896061
]
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice>
$env:SERVICE_MODULE="serve_churn"
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose up
-d --force-recreate api
```

```
[+] up 1/1
```

```
✔ Container cartvista-lab3-api-1 Started 0.6s
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
cartvista-lab3-api-1	cartvista-dev:lab	"uvicorn serve_churn..."	api	10 seconds ago	Up 8 seconds	127.0.0.1:8000->8000/tcp

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
logs --tail 100 api
```

```
api-1 | INFO: Started server process [1]
```

```
api-1 | INFO: Waiting for application startup.
```

```
api-1 | INFO: Application startup complete.
```

```
api-1 | INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm tools python probe_service.py --url http://api:8000 | Tee-Object -FilePath
.\Lab3_Submission\Prediction_Tests.txt
```

Container cartvista-lab3-tools-run-c09d2ef33dd0 Creating

Container cartvista-lab3-tools-run-c09d2ef33dd0 Created

out-file : Could not find a part of the path

'C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice\Lab3_Submission\Prediction_Tests.txt'.

At line:1 char:1

```
+ docker compose run --rm tools python probe_service.py --url http://ap ...
```

```
+ ~~~~~
```

```
+ CategoryInfo          : OpenError: (:) [out-file], DirectoryNotFoundException
```

```
+ FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm tools python probe_service.py --url http://api:8000 | Tee-Object -FilePath
.\Lab3_Model_Serving_Microservice\Prediction_Tests.txt
```

Container cartvista-lab3-tools-run-7ea3f49ccdd0 Creating

Container cartvista-lab3-tools-run-7ea3f49ccdd0 Created

out-file : Could not find a part of the path

'C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice\Lab3_Model_Serving_Microservice\Prediction_Tests.txt'.

At line:1 char:1

```
+ docker compose run --rm tools python probe_service.py --url http://ap ...
```

```
+ ~~~~~
```

```
+ CategoryInfo          : OpenError: (:) [out-file], DirectoryNotFoundException
```

```
+ FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm tools python probe_service.py --url http://api:8000 | Tee-Object -FilePath
.\Lab3_Submission\Prediction_Tests.txt
```

Container cartvista-lab3-tools-run-054cc8724dab Creating

Container cartvista-lab3-tools-run-054cc8724dab Created

{

```
"health": {
  "status": 200,
  "body": {
    "status": "ok",
    "model_loaded": true
  }
},
"model": {
  "status": 200,
  "body": {
    "model": "cartvista-churn",
    "alias": "champion",
    "version": 2,
    "run_id": "f0dc4382ba9840789fd4d27177ad45a5"
  }
},
"at_risk": {
  "status": 200,
  "body": {
    "churn_prediction": 1,
    "churn_probability": 0.8805979175205052,
    "model_version": 2,
    "inference_ms": 1.953
  }
},
"loyal": {
  "status": 200,
  "body": {
    "churn_prediction": 0,
    "churn_probability": 0.002663625308117632,
    "model_version": 2,
```

```
    "inference_ms": 1.311
  }
},
"invalid": {
  "status": 422,
  "body": {
    "detail": [
      {
        "type": "greater_than_equal",
        "loc": [
          "body",
          "tenure_months"
        ],
        "msg": "Input should be greater than or equal to 0",
        "input": -5,
        "ctx": {
          "ge": 0
        }
      }
    ]
  }
},
"empty_batch": {
  "status": 422,
  "body": {
    "detail": [
      {
        "type": "too_short",
        "loc": [
          "body",
          "customers"
        ]
      }
    ]
  }
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> Invoke-
RestMethod http://localhost:8000/model-info | ConvertTo-Json | Tee-Object -FilePath
.\Lab3_Submission\Model_Before.json
```

```
{
  "model": "cartvista-churn",
  "alias": "champion",
  "version": 2,
  "run_id": "f0dc4382ba9840789fd4d27177ad45a5"
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
run --rm tools python register_model.py
```

Container cartvista-lab3-tools-run-dc1aa0cf72bd Creating

Container cartvista-lab3-tools-run-dc1aa0cf72bd Created

Registered model 'cartvista-churn' already exists. Creating a new version of this model...

Created version '3' of model 'cartvista-churn'.

```
{
  "alias": "champion",
  "version": 3,
  "run_id": "9d56073a3f454c8aba3d10fe61d999b1",
  "test_auc": 0.7495794388998274
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> Invoke-
RestMethod http://localhost:8000/model-info | ConvertTo-Json | Tee-Object -FilePath
.\Lab3_Submission\Model_After_Alias_Move.json
```

```
{
  "model": "cartvista-churn",
  "alias": "champion",
  "version": 2,
  "run_id": "f0dc4382ba9840789fd4d27177ad45a5"
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
restart api
```

[+] restart 0/1

- Container cartvista-lab3-api-1 Restarting

3.6s

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> Invoke-
RestMethod http://localhost:8000/model-info | ConvertTo-Json | Tee-Object -FilePath
.\Lab3_Submission\Model_After_Restart.json
```

```
{
  "model": "cartvista-churn",
  "alias": "champion",
  "version": 3,
  "run_id": "9d56073a3f454c8aba3d10fe61d999b1"
}
```

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> python labctl.py
export
```

Container cartvista-lab3-tools-run-133245439d1b Creating

Container cartvista-lab3-tools-run-133245439d1b Created

Exported 3 to /workspace/model_export

Container cartvista-lab3-copy Creating

Container cartvista-lab3-copy Created

d20a22353403d3b8c30f0b9bb9b13e962f3430b7430ebcee7bce26ab02029fc3

Successfully copied 18.9kB to

C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice\model_export

cartvista-lab3-copy

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker build -f
Dockerfile.service -t cartvista-api:v1 .
```

```
[+] Building 81.1s (12/12) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile.service        0.0s
=> => transferring dockerfile: 382B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim
1.6s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 105B                                    0.0s
=> [1/6] FROM docker.io/library/python:3.11-slim@sha256:e41613d42d4891e4930f79523f93f81bbc7632584ec65e36ab055f41a800b41e  0.0s
=> [internal] load build context                                  0.0s
```

```

=> => transferring context: 15.53kB                                0.0s
=> CACHED [2/6] WORKDIR /app                                       0.0s
=> [3/6] COPY requirements.txt /tmp/requirements.txt
0.0s
=> [4/6] RUN pip install --no-cache-dir -r /tmp/requirements.txt
74.6s
=> [5/6] COPY model_export/ model_export/                          0.1s
=> [6/6] COPY api.py common.py drift.py serve_churn.py ./
0.1s
=> exporting to image                                              4.4s
=> => exporting layers                                             4.4s
=> => writing image
sha256:e36de15781c79b1a4817b0ddd8d8bfa52b59521650e8b00330b7b1dee69014f5
0.0s
=> => naming to docker.io/library/cartvista-api:v1                0.0s

```

What's Next?

View a summary of image vulnerabilities and recommendations → [docker scout quickview](#)

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker images
cartvista-api
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
cartvista-api	v1	e36de15781c7	21 seconds ago	926MB

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose
stop api
```

```
[+] stop 1/1
```

✓ Container cartvista-lab3-api-1 Stopped 5.1s

```
PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker run --rm -p
127.0.0.1:8000:8000 cartvista-api:v1
```

```
INFO: Started server process [1]
```

```
INFO: Waiting for application startup.
```

```
INFO: Application startup complete.
```

```
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

```
INFO: 172.17.0.1:42474 - "GET /health HTTP/1.1" 200 OK
```

```
INFO: 172.17.0.1:42474 - "GET /favicon.ico HTTP/1.1" 404 Not Found
```

INFO: 172.17.0.1:42476 - "GET /model-info HTTP/1.1" 200 OK
INFO: 172.17.0.1:42486 - "GET /docs HTTP/1.1" 200 OK
INFO: 172.17.0.1:42486 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 172.17.0.1:58126 - "POST /predict HTTP/1.1" 200 OK
INFO: 172.17.0.1:58126 - "POST /predict HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]

Aborted!

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose stop api mlflow

[+] stop 1/1

✔ Container cartvista-lab3-api-1 Stopped 0.0s

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose run --rm --no-deps tools python -c "from pathlib import Path; p=Path('mlflow.db'); b=Path('mlflow.db.lab3-backup'); assert p.exists() and not b.exists(), 'Check database and backup before proceeding'; p.rename(b); print('Registry temporarily renamed')"

Container cartvista-lab3-tools-run-2c84556e9875 Creating

Container cartvista-lab3-tools-run-2c84556e9875 Created

Registry temporarily renamed

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose up -d api

[+] up 1/1

✔ Container cartvista-lab3-api-1 Started 0.5s

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose stop api

[+] stop 1/1

✔ Container cartvista-lab3-api-1 Stopped 0.8s

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose run --rm --no-deps tools python -c "from pathlib import Path; p=Path('mlflow.db'); b=Path('mlflow.db.lab3-backup'); assert b.exists() and not p.exists(), 'Check files before restoring'; b.rename(p); print('Registry restored')"

Container cartvista-lab3-tools-run-77a412cab1e2 Creating

Container cartvista-lab3-tools-run-77a412cab1e2 Created

Registry restored

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> docker compose up
-d api

[+] up 1/1

✓ Container cartvista-lab3-api-1 Started 0.4s

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice> python labctl.py
collect

Container cartvista-lab3-copy Creating

Container cartvista-lab3-copy Created

1ba025f3c13ddae76c230b5b0ec9bde19452bce42b76af76e060d5929e2e848a

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ort

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.csv

cartvista-lab3-copy

Saved evidence in

C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice\my_results

PS C:\Users\saian\Downloads\MLOps_Labs\Lab3_Model_Serving_Microservice>

The screenshot shows a terminal window with a Curl command and its output. The command is: `curl -X 'GET' \ 'http://localhost:8000/health' \ -H 'accept: application/json' \`. The response is a JSON object: `{ "status": "ok", "model_loaded": true }`. The response headers are: `content-length: 35, content-type: application/json, date: Mon, 28 Sep 2026 11:22:03 GMT, server: uvicorn`.

Code	Details
200	Response body <pre>{ "status": "ok", "model_loaded": true }</pre> Response headers <pre>content-length: 35 content-type: application/json date: Mon, 28 Sep 2026 11:22:03 GMT server: uvicorn</pre>

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "tenure_months": -5,
    "monthly_spend": 2500.0,
    "orders_per_month": 1,
    "support_tickets": 6,
    "uses_discounts": 0,
    "app_sessions": 2
  }'
```

Request URL

http://localhost:8000/predict

Server response

Code	Details
------	---------

422	Error: Unprocessable Entity
-----	-----------------------------

Response body

```
{
  "detail": [
    {
      "type": "greater_than_equal",
      "loc": [
        "body",
        "tenure_months"
      ],
      "msg": "Input should be greater than or equal to 0",
      "input": -5,
      "ctx": {
        "ge": 0
      }
    }
  ]
}
```

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "tenure_months": 60,
    "monthly_spend": 1500.0,
    "orders_per_month": 8,
    "support_tickets": 0,
    "uses_discounts": 1,
    "app_sessions": 24
  }'
```

Request URL

http://localhost:8000/predict

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
{
  "churn_prediction": 0,
  "churn_probability": 0.002663625308117632,
  "model_version": 2,
  "inference_ms": 1.914
}
```

Curl

```

curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "tenure_months": 2,
    "monthly_spend": 2500.0,
    "orders_per_month": 1,
    "support_tickets": 6,
    "uses_discounts": 0,
    "app_sessions": 2
  }'

```

Request URL

```
http://localhost:8000/predict
```

Server response

Code	Details
200	Response body <pre> { "churn_prediction": 1, "churn_probability": 0.8805979175205052, "model_version": 2, "inference_ms": 1.458 } </pre> Download

```

{
  "health": {
    "status": 200,
    "body": {
      "status": "ok",
      "model_loaded": true
    }
  },
  "model": {
    "status": 200,
    "body": {
      "model": "cartvista-churn",
      "alias": "champion",
      "version": 2,
      "run_id": "f0dc4382ba9840789fd4d27177ad45a5"
    }
  },
  "at_risk": {
    "status": 200,
    "body": {
      "churn_prediction": 1,
      "churn_probability": 0.8805979175205052,
      "model_version": 2,

```

```
    "inference_ms": 1.953
  }
},
"loyal": {
  "status": 200,
  "body": {
    "churn_prediction": 0,
    "churn_probability": 0.002663625308117632,
    "model_version": 2,
    "inference_ms": 1.311
  }
},
"invalid": {
  "status": 422,
  "body": {
    "detail": [
      {
        "type": "greater_than_equal",
        "loc": [
          "body",
          "tenure_months"
        ],
        "msg": "Input should be greater than or equal to 0",
        "input": -5,
        "ctx": {
          "ge": 0
        }
      }
    ]
  }
},
```

```

"empty_batch": {
  "status": 422,
  "body": {
    "detail": [
      {
        "type": "too_short",
        "loc": [
          "body",
          "customers"
        ],
        "msg": "List should have at least 1 item after validation, not 0",
        "input": [],
        "ctx": {
          "field_type": "List",
          "min_length": 1,
          "actual_length": 0
        }
      }
    ]
  }
}

```

POST

/predict Predict

^

Parameters

Cancel

Reset

No parameters

Request body required

application/json

▼

```

{
  "tenure_months": 2,
  "monthly_spend": 2500.0,
  "orders_per_month": 1,
  "support_tickets": 6,
  "uses_discounts": 0,
  "app_sessions": 2
}

```

}